



Application of algae for heavy metal adsorption: A 20-year meta-analysis

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ABSTRACT

The use of algae to adsorb heavy metals is an efficient and environmentally friendly treatment for contaminated water and has attracted widespread research attention. In this study, a meta-analysis of the heavy metal adsorption capacity of algae from five different phyla and the factors influencing these capacities was conducted. Phaeophyta was found to have a high heavy metal adsorption capacity, whereas Bacillariophyta had a relatively low adsorption capacity; Chlorophyta, Rhodophyta, and Cyanophyta had moderate adsorption capacities. Non-living algae were more effective in practical applications than living algae were. Algal biomass had a relatively high adsorption efficiency of 1–10 g/L, which did not increase significantly when algal concentration increased. The algal adsorption efficiency for initial heavy metal concentrations of 10–100 mg/L was higher than for concentrations of greater than 100 mg/L. The results further show that algal adsorption of heavy metals reached a maximum capacity of 80–90% within 20 min. Heavy metal adsorption by algae was not temperature-dependent, and it was more effective in moderately to weakly acidic environments (pH = 4–7.5). Considering these aspects for practical applications, algae from some phyla can effectively be used for heavy metal biosorption in contaminated water.

1. Introduction

The pollution of water with heavy metals is a critical global problem (Meybeck and Helmer, 1989; Järup, 2003; Rickwood and Carr, 2009; Fernandez-Luqueno et al., 2013; Islam et al., 2015). With the development of mining, smelting, and other industrial techniques, large quantities of heavy metal-contaminated water are being discharged into the environment (Nagajyoti et al., 2010; Callender, 2014). Highly toxic and persistent accumulated industrial wastes often contain large quantities of heavy metals. Heavy metals accumulate in the environment through exhaust gas, sedimentation, or runoff (Leung and Jiao, 2006; Hashim et al., 2011; Galitskaya et al., 2017) or through direct discharge of industrial wastewater and domestic sewage (Hiller et al., 2010; Islam et al., 2015). This results in excessive heavy metal concentrations in water bodies, which usually does not exceed 5 mg/L in natural rivers globally (Gaillardet et al., 2003). Heavy metal concentrations in wastewater can reach hundreds of milligrams per liter (Fernandes et al., 2007; Rahmanpour et al., 2014), which is far beyond the threshold acceptable for human use (Ho, 1990; Hiller et al., 2010; Islam et al., 2015; Wang et al., 2016a).

Heavy metals are trace elements (such as Cd, Cr, Cu, Pb, Zn, Co, Ni, and Hg) that are toxic to organisms at certain concentrations. For Cd, a concentration higher than 1 mg/kg will cause damage; for Cr, damage

occurs at concentrations above 50 µg per liter (Duffus, 2002; Appenroth, 2010; Tchounwou et al., 2012). They can accumulate in the environment via physical, chemical, and physiological processes and may have toxic effects on organisms, ecosystems, and humans via the food chain (Järup, 2003; Sciacca and Conti, 2009; Tchounwou et al., 2012; Avigliano et al., 2015; Dadar et al., 2016; Fakhri et al., 2018). Furthermore, some heavy metals may seriously threaten humans through direct exposure to contaminated air, soil, or water (Järup, 2003). If not rapidly treated, toxic heavy metals may become persistent in the environment (Bergomi et al., 1997; Callender, 2003, 2014; Copat et al., 2014; Bo et al., 2015; Ferrante and Conti, 2017; Saleh et al., 2019). Hence, the use appropriate of water treatment techniques is imperative to decrease heavy metal concentrations in wastewaters, for example, through coagulation with precipitation, ion exchange, adsorption, or membrane technology (Lone et al., 2008). Methods such as ion exchange are relatively rapid but often cause secondary environmental pollution or are limited to treating wastewaters with relatively low heavy metal concentrations (mostly less than 100 mg/L) (Fu and Wang, 2011).

Bioremediation methods are widely used for their effectiveness, low cost, and provision of environmental protection. In particular, biosorption is the process of heavy metal sequestration by non-living biomass from algae, lichens, mushrooms, mosses, bacteria, and biomass

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waste, including rice husks, wood fibers, sawdust, fly ash, and tea stems (Kratochvil and Volesky, 1998; Çetinkaya Dönmez et al., 1999; Chuah et al., 2005; Uluzlu et al., 2010; Hegazi, 2013; Abdolali et al., 2016; Kariuki et al., 2017). Various living organisms, such as yeast, bacteria, aquatic plants, and algae, can also be used for biosorption (Gadd, 1990; Çetinkaya Dönmez et al., 1999; Keskinan et al., 2004; Mishra and Tripathi, 2008; Ahemad, 2014; Amirnia et al., 2015; Rezaia et al., 2016). The adsorption of heavy metals in polluted water by aquatic plants and algae has received considerable research attention because of its high efficiency and wide range of applications (Keskinan et al., 2003). Heavy metal adsorption is a cost-effective means of achieving environmental protection, especially because algae reproduce rapidly (Sandau et al., 1996; Wang et al., 2016b; Zeraatkar et al., 2016; Mohd Udaiyappan et al., 2017). Under experimental conditions, the adsorption rate of heavy metals by most macro- and microalgae can reach more than 80% and, in some cases, up to 95% (Lau et al., 2003; Ghimire et al., 2008). Living algae applied to a suitable place (such as contaminated water that is not polluted enough to poison it) can reproduce rapidly, thereby efficiently adsorbing heavy metal ions in the polluted water body and reducing treatment costs. Non-living algae can yield a large number of products to treat pollution with a small amount of raw materials and has great industrial production value and development prospects (Abdolali et al., 2016).

In this study, a meta-analysis was carried out to determine the differences in algal adsorption capacity for different heavy metals as well as the optimum adsorption temperature, pH, time, heavy metal concentration, algal concentration, and algal activity during the adsorption of heavy metals. The advantages and disadvantages of algae in sewage treatment were identified to determine the optimal physical and chemical environment and dosage required for treating heavy metal-contaminated water with algae.

2. Methods

2.1. Document retrieval

This meta-analysis was carried out according to the Preferred Reporting Items for Systematic Reviews and Meta-analyses (Moher et al., 2015) guidelines. The study selection is illustrated in appendix A. The English literature search was performed using Elsevier, Wiley, Springer, PubMed and Taylor & Francis databases, and the Chinese literature search was conducted via the CNKI database. In our search strategy, we used Chinese and English keyword combinations of “algae” and “heavy metal” and obtained a total of 353 potential articles published between 1998 and 2018. Of the 353 articles retrieved, 60 duplicates were deleted. The title and abstract of the remaining 293 articles were then reviewed, which led to 140 being deemed unsuitable (not experimental articles or different research methods). After reviewing the data of the remaining 153 articles, 96 that did not focus on algae or lacked standard deviations were omitted. The text of the remaining 57 articles was reviewed (the amount of algae is unknown or the experimental data is incomplete), and 28 articles, which were missing crucial factors (temperature, pH, or time) and had unreliable data, were further omitted. Finally, a total of 29 eligible articles were used for the meta-analysis. These articles reported both sufficient experimental details (algal species, metal type, algal status, biomass) and environmental factors (initial heavy metal concentration, time, temperature, pH) that reflected the ability of algae to adsorb heavy metal ions.

2.2. Statistical analyses

The meta-analysis was conducted using OpenMEE 8.1 software in the R Metafor package. In accordance with the principles of integrated analysis, each article was considered as independent, with different pH values, initial heavy metal concentrations, algal source areas, and use of

either living or non-living algae. Based on the data reported in each article, transformation analyses were performed to compare the abilities of different algae to adsorb heavy metal ions.

To eliminate differences caused by factors such as heavy metal concentrations in different experiments and varying algal adsorption efficiencies, the reaction ratio $\ln R$ was used as the Effect Size, and was calculated using the following formula:

$$\ln R = \ln(X_e/X_c) \quad (1)$$

where X_e is the concentration of heavy metal ions adsorbed by algae (i.e., the initial heavy metal concentration in water minus the concentration of residual heavy metals in the water after adsorption by algae), and X_c is the heavy metal concentration in surrounding water (Chen et al., 2018). This comprehensive parameter has been used in multiple studies and was originally proposed by Hedges (Hedges et al., 1999). The larger the $\ln R$ value is, the stronger is the ability of algae to adsorb heavy metal ions from water, and the higher is the adsorption efficiency. Determination of the mean and 95% confidence interval (CI) of $\ln R$ was performed using OpenMEE 8.1 software (Zvereva and Kozlov, 2014; Benitez-Lopez et al., 2017).

In meta-analyses, heterogeneity is usually used as a basis for evaluating the significance of differences between studies and can be estimated as a weighted sum of squares, called the Q statistic. Heterogeneity testing can be performed by comparing Q_T to a χ^2 distribution with K-1 degrees of freedom (Borenstein et al., 2011; Koricheva et al., 2013; Tozser et al., 2017). When $P < 0.05$, the differences are considered significant (Wang et al., 2015). In meta-analyses, the significance is usually determined by the 95% CI (Borenstein et al., 2011; Zvereva and Kozlov, 2014; Li et al., 2015; Rossetti et al., 2017). The processed data are listed in Table 1.

3. Results and discussion

3.1. Metal type

A total of 117 points of data analysis of seven different heavy metals (Cd, Cr, Hg, Co, Pb, Cu, and Zn) were included in this study. Cd had 37 data points, Cr^{3+} had 10, Cr^{6+} had seven, Hg had four, Co had six, Pb had 17, Cu had 26, and Zn had 10 data points. The algal adsorption capacity differed significantly for the different metals (Fig. 1(a)). The ES values of Cd^{2+} , Cr^{3+} , Co^{2+} , Zn^{2+} , and Pb^{2+} were between -1.0 and -0.4 , and those of Hg^{2+} , Cr^{6+} , and Cu^{2+} were lower than -1.0 . The Hg^{2+} CI was the highest, whereas that of Zn^{2+} was the lowest. A relatively large number of data was used for Cd^{2+} and Cu^{2+} , and the small standard error indicated the reliability of the data. Algae had a high adsorption capacity for metal ions such as Cd^{2+} , Cr^{3+} , Co^{2+} , Pb^{2+} , and Zn^{2+} , a moderate adsorption capacity for Cu^{2+} and Cr^{6+} , and a low adsorption capacity for Hg^{2+} .

According to previous research, algae can adsorb metal ions via two main mechanisms, one of which is related to the coordination, complexation, ion exchange, physical adsorption, and microprecipitation of metal ions. This mechanism (namely bioaccumulation) is independent of physiological factors and involves only algal cell structure and surface groups (Davis et al., 2003; Zhou and Qiu, 2004). The other type of metal adsorption (namely, biosorption) is related to algal metabolism, during which metal ions are stored within the algal body (Mehta and Gaur, 2008; Wang and Chen, 2009; Rajfur, 2013; Anastopoulos and Kyzas, 2015; Suresh Kumar et al., 2015). Biosorption mainly involves surface complexation, ion exchange, and redox reactions.

Surface complexation mainly involves the combination of heavy metal ions in solution and organic functional groups on the algal surface. In related studies, the surface microstructure of algae following the adsorption of Cu^{2+} ions is sharper (Chen and Yang, 2005), indicating that heavy metal ions have been complexed to the algal surface. During complexation, heavier ions are more suitable for binding to sites with two distal functional groups. Therefore, Pb^{2+} , Cu^{2+} , and

Table 1
Effect size and references used in this meta-analysis.

Phylum	Algal Species	LnR									Reference
		Zn	Pb	Ni	Hg	Cu	Cr ⁶⁺	Cr ³⁺	Co	Cd	
Bacillariophyta	<i>Cyclotella cryptica</i>		−1.554		−1.814	−1.86				−1.924	Schmitt et al., (2001)
	<i>Phaeodactylum tricornutum</i>		−1.997		−2.087	−1.836				−2.315	Schmitt et al., (2001)
	<i>Chaetoceros</i> sp.									−0.01	Wang et al., (2017)
	<i>Nitzschia</i> sp.									−0.028	Wang et al., (2017)
Chlorophyta	<i>Thalassiosira</i> sp.									−0.04	Wang et al., (2017)
	<i>Scenedesmus obliquus</i> CNW-N									−0.023	Chen et al., (2012)
	<i>Chlamydomonas microspheara</i>									−0.27	Jiang et al., (2018)
	<i>Scenedesmus quadricauda</i>									−2.313	Jiang et al., (2018)
	<i>Ulva fasciata</i>					−0.029					Karthikeyan et al., (2007)
	<i>Ulva lactuca</i>						−1.504	−1.21			Murphy et al., (2008)
	<i>Ulva</i> spp.						−1.41	−0.849			Murphy et al., (2008)
	<i>Ulva lactuca</i>					−2.434					Murphy et al., (2009)
	<i>Ulva</i> spp.					−2.364					Murphy et al., (2009)
	<i>Caulerpa lentillifera</i>	−0.116	−0.091			−0.167				−0.213	Pavasant et al., (2006)
	<i>Chlorella vulgaris</i>			−0.984		−0.856					Rangsayatorn et al., (2004)
	<i>Scenedesmus subspicatus</i>		−0.281		−1.016	−1.399				−1.951	Schmitt et al., (2001)
	<i>Chlorella minutissima</i> UTEX2341 (freeze) ^a	−0.031				−0.151				−0.122	Yang et al., (2015)
	<i>Chlorella minutissima</i> UTEX2341 (light) ^a	−0.477				−0.179				−0.297	Yang et al., (2015)
Cyanophyta	<i>Arthrospira platensis</i> (dried) ^a					−0.099					Al-Homaidan et al., (2014)
	<i>Synechocystis</i> sp. PCC 6803		−0.442							−0.763	Arunakumara, (2007)
	<i>Merismopedia tenuissima</i>					−0.654				−0.405	Fawzy, (2016)
	<i>Arthrospira platensis</i> (feed1600lx) ^a									−1.879	Jiang et al., (2018)
	<i>Microcystis aeruginosa</i>									−1.569	Jiang et al., (2018)
	<i>Arthrospira platensis</i>	−0.949							−1.234	−0.042	Rangsayatorn et al., (2004)
	<i>Arthrospira platensis</i> (freeze-drying) ^a					−0.248					Vannela and Verma, (2006)
	<i>Arthrospira platensis</i> (feed4000lx) ^a									−4.062	Wang et al., (2007)
	<i>Spirulina maxima</i>									−4.522	Wang et al., (2007)
	<i>Microcystis</i>			−0.59		−0.33				−0.155	Wu et al., (2014)
Phaeophyta	<i>Sargassum</i> sp.									−0.014	Cruz et al., (2004)
	<i>Sargassum flipendula</i> 1					−0.666				−0.662	Davis et al., (2000)
	<i>Sargassum fluitans</i>					−0.754				−0.542	Davis et al., (2000)
	<i>Sargassum vulgare</i>					−0.546				−0.462	Davis et al., (2000)
	<i>Bifurcaria bifurcata</i>	−0.553	−0.458							−0.269	Freitas et al., (2008)
	<i>Laminaria hyperborea</i>	−0.832	−0.06							−0.363	Freitas et al., (2008)
	<i>Sargassum muticum</i>	−0.452	−0.343								Freitas et al., (2008)
	<i>Sargassum muticum</i> (20) ^a									−0.255	Freitas et al., (2008)
	<i>Sargassum myriocystum</i>						−0.698				Jayakumar et al., (2015)
	<i>Sargassum</i> sp.					−0.038					Karthikeyan et al., (2007)
	<i>Sargassum thunbergii</i>	−0.207									Li et al., (2017)
	<i>Sargassum muticum</i> (60) ^a									−0.24	Lodeiro et al., (2004)
	<i>Cystoseira baccata</i>		−0.119							−0.47	Lodeiro et al., (2006)
	<i>Durvillaea potatorum</i> DP95Ca		−0.008			−0.101					Matheickal and Yu, (1999)
	<i>Ecklonia radiata</i> ER95Ca		−0.052			−0.068					Matheickal and Yu, (1999)
	<i>Fucus spiralis</i>	−0.445	−0.172				−1.253	−0.708		−0.173	Murphy et al., (2008)
	<i>Fucus vesiculosus</i>						−1.063	−0.672			Murphy et al., (2008)
	<i>Fucus spiralis</i>					−2.157					Murphy et al., (2009)
	<i>Fucus vesiculosus</i>					−2.043					Murphy et al., (2009)
Rhodophyta	<i>Sargassum</i> sp.								−1.315		Soleymani et al., (2015)
	<i>Gracilaria fisher</i>					−0.329				−0.462	Chaisuksant (2003)
	<i>Corallina mediterranea</i>		−0.181					−0.372	−0.291	−0.468	Ibrahim (2011)
	<i>Galaxaura oblongata</i>		−0.109					−0.193	−0.545	−0.382	Ibrahim (2011)
	<i>Jania rubens</i>		−0.301					−0.497	−0.363	−0.427	Ibrahim (2011)
	<i>Pterodocladia capillacea</i>		−0.285					−0.519	−0.102	−0.543	Ibrahim (2011)
	<i>Palmaria palmata</i>						−1.304	−1.399			Murphy et al., (2008)
	<i>Polysiphonia lanosa</i>						−0.999	−1.335			Murphy et al., (2008)
	<i>Palmaria palmata</i>					−2.907					Murphy et al., (2009)
	<i>Polysiphonia lanosa</i>					−2.557					Murphy et al., (2009)
	<i>Hypnea valentiae</i>									−0.146	Rathinam et al., (2010)
	<i>Porphyridium purpureum</i>	−0.223	−2.132		−1.168	−2.292				−1.896	Schmitt et al., (2001)

^a Different treatments in preprocessing and experimentation, where “20” and “60” respectively indicate 20 and 60 °C; “light” indicates that algae were not lyophilized after having been cultivated in light; “freeze” indicates that algae were freeze-dried after having been cultivated in light; “feed1600lx” means algae were cultured under a light intensity of 1600 lx; “feed4000lx” means algae were cultured under a light intensity of 4000 lx; “dried” means dry after grinding treatment, “freeze-drying” means artificial culture and freeze-drying treatment.

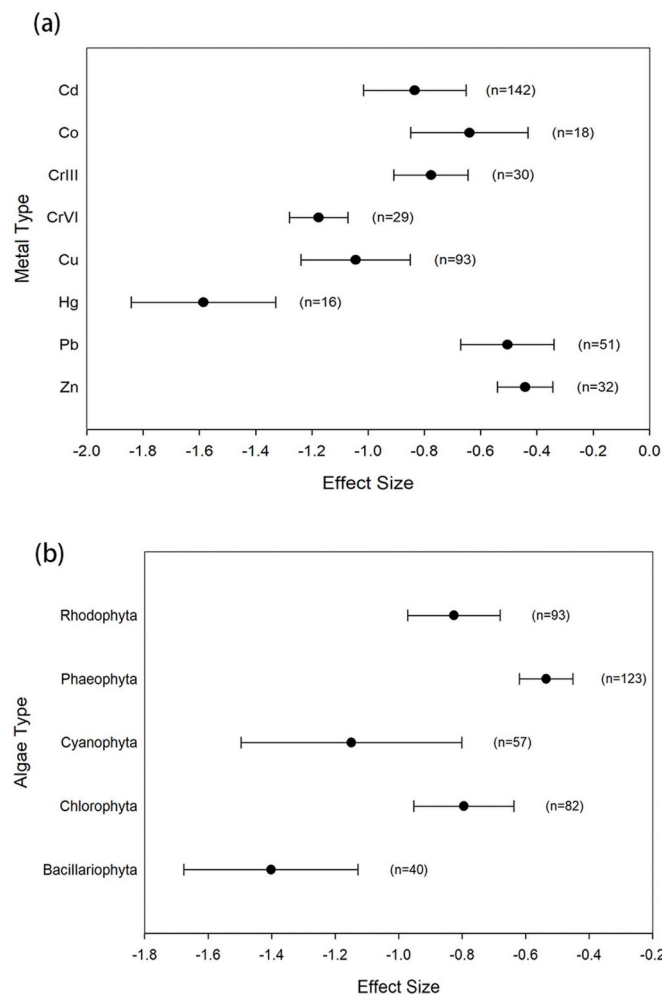


Fig. 1. (a) Difference in effect size of algal adsorption of different heavy metal ions. (b) Influence of different algal phyla on heavy metal adsorption. Effect size was processed by Ln, and error bars represent $\pm$ 95% confidence interval; n represents number of samples for each indicator. Data were tested using heterogeneity (Q), $P < 0.05$.

Cd^{2+} ions are first in the affinity sequence when complexed (Davis et al., 2003), which is consistent with Fig. 1(a). Ion exchange usually involves a light metal ion carried on the algal surface and exchanged with heavy metal ions in solution. Following adsorption, some metal cations are not present in the solution before the process (Chojnacka et al., 2005). Redox reactions can reduce Cr^{6+} to Cr^{3+} and diminish its toxicity (Yang and Chen, 2008). In shown in Fig. 1(a), the adsorption capacity of Zn^{2+} and Pb^{2+} is relatively high, indicating that the removal of heavy metal ions may occur via biosorption and bioconcentration.

3.2. Algal species

The algal species had a certain influence on heavy metal adsorption. However, in this study, the collected data were insufficient to identify algae to species level, and algae were identified to phylum level. A total of 119 datasets was collected and five algal phyla were identified. Four phyla (Phaeophyta, Chlorophyta, Rhodophyta, and Cyanophyta) exhibited relatively high adsorption efficiencies, but Bacillariophyta had a relatively low adsorption efficiency (Fig. 1(b)). In addition, the ES values for Chlorophyta (-0.79), Rhodophyta (-0.83), and Phaeophyta (-0.54) were greater than those for Cyanobacteria (-1.15) and Bacillariophyta (-1.40), indicating higher adsorption efficiencies. More data were available for Chlorophyta ($n = 82$), Rhodophyta ($n = 93$),

and Phaeophyta ($n = 123$) than for Cyanophyta ($n = 57$) and Bacillariophyta ($n = 40$). The 95% CI of Rhodophyta and Phaeophyta were low, indicating high data reliability.

Bacillariophyta had the lowest heavy metal adsorption efficiency (Fig. 1(b)), possibly due to their structural characteristics. The cell wall of diatoms in this phylum is mainly composed of two flaps called frustules, which are siliceous and contain pectin, and their photosynthetic products mainly consist of lipids (Mann et al., 2016). During biosorption and bioaccumulation (when algae adsorb heavy metals), few exposed parts of the diatoms undergo exchange or complexation with heavy metals because of the frustules. Bacillariophyta are also not suitable for binding heavy metals because of the structure of the photosynthetically produced lipids. Therefore, these diatoms have limited practical applicability in treating heavy metal-contaminated water.

Fig. 1(b) shows that Phaeophyta exhibited considerably higher heavy metal adsorption performance than other algal phyla did. As a multi-cellular plant body, the surface layer of algae in this phylum has a large contact area with the surrounding water, which is favorable for heavy metal adsorption. Furthermore, fucoidan in the cell walls forms mucus on the plant surface and promotes the adsorption of heavy metals through complexation with polysaccharides. Phaeophyta also contain a small and unique acidic vacuole that provides a suitable environment for heavy metal binding (Davis et al., 2003).

3.3. Heavy metal types and algal species

This study identified differences in the adsorption capacity of algal species for different heavy metals. Algal species were grouped according to the type of heavy metal adsorbed. A total of 94 groups of data were made with seven heavy metal ions and five algal phyla. Different algal phyla had different adsorption capacities for the same heavy metal ion. For example, the ES values for Zn^{2+} adsorption of Chlorophyta and Phaeophyta were -0.21 and -0.50 , respectively. The order of Cu^{2+} adsorption efficiency was Cyanophyta ($\text{ES} = -0.23$) > Phaeophyta ($\text{ES} = -0.80$) > Chlorophyta ($\text{ES} = -0.95$) > Rhodophyta ($\text{ES} = -2.02$). The Pb^{2+} adsorption efficiency of Phaeophyta was highest ($\text{ES} = -0.17$), and the Cr^{6+} adsorption efficiency was lowest ($\text{ES} = -1.01$). The Cd^{2+} adsorption efficiency of Phaeophyta was highest ($\text{ES} = -0.34$), and those of Chlorophyta ($\text{ES} = -0.74$) and Rhodophyta ($\text{ES} = -0.61$) were similar. In summary, the adsorption efficiency of algae from different phyla differs depending on the metal.

In general, Phaeophyta had the highest ES value (-0.54) and adsorption efficiency, and the low 95% CI (Figs. 1 and 2) indicated that these data had the highest credibility. This is consistent with previous research (Volesky and Holan, 1995), indicating that the adsorption performance of Phaeophyta for different metals was consistent and stable, with high practical value. In general, the adsorption efficiency of different heavy metals by Phaeophyta could be ranked as $\text{Pb}^{2+} > \text{Cd}^{2+} > \text{Zn}^{2+} > \text{Cu}^{2+} > \text{Cr}^{6+}$ (Davis et al., 2003; Romera et al., 2007; He and Chen, 2014; Anastopoulos and Kyzas, 2015).

Furthermore, the use of other algae (Chlorophyta, Rhodophyta, and Cyanophyta) to treat heavy metal-contaminated water (Cr^{3+} , Co^{2+} , Cu^{2+} , and Zn^{2+}) required targeted selection based on the environment parameters and usage requirements (Figs. 1 and 2). Chlorophyta ($\text{ES} = -0.79 \pm 0.16$; 95% CI) can generally adsorb Cd^{2+} , and the high 95% CI indicates that the adsorption efficiency varies greatly. In addition, the selectivity of this phylum to heavy metals is higher, i.e., the ability to adsorb various heavy metals differs. Specifically, the order of adsorption performance is Zn^{2+} ($\text{ES} = -0.21$) > Cd^{2+} ($\text{ES} = -0.74$) > Cu^{2+} ($\text{ES} = -0.95$). The adsorption efficiency of Rhodophyta ($\text{ES} = -0.83 \pm 0.15$; 95% CI) was unstable, because the species in this phylum had different performances in different environments and thus need to be pre-selected for practical use. The adsorption efficiency of Cyanophyta ($\text{ES} = -1.15 \pm 0.35$; 95% CI) was low, and species from this phylum are rarely used in practical

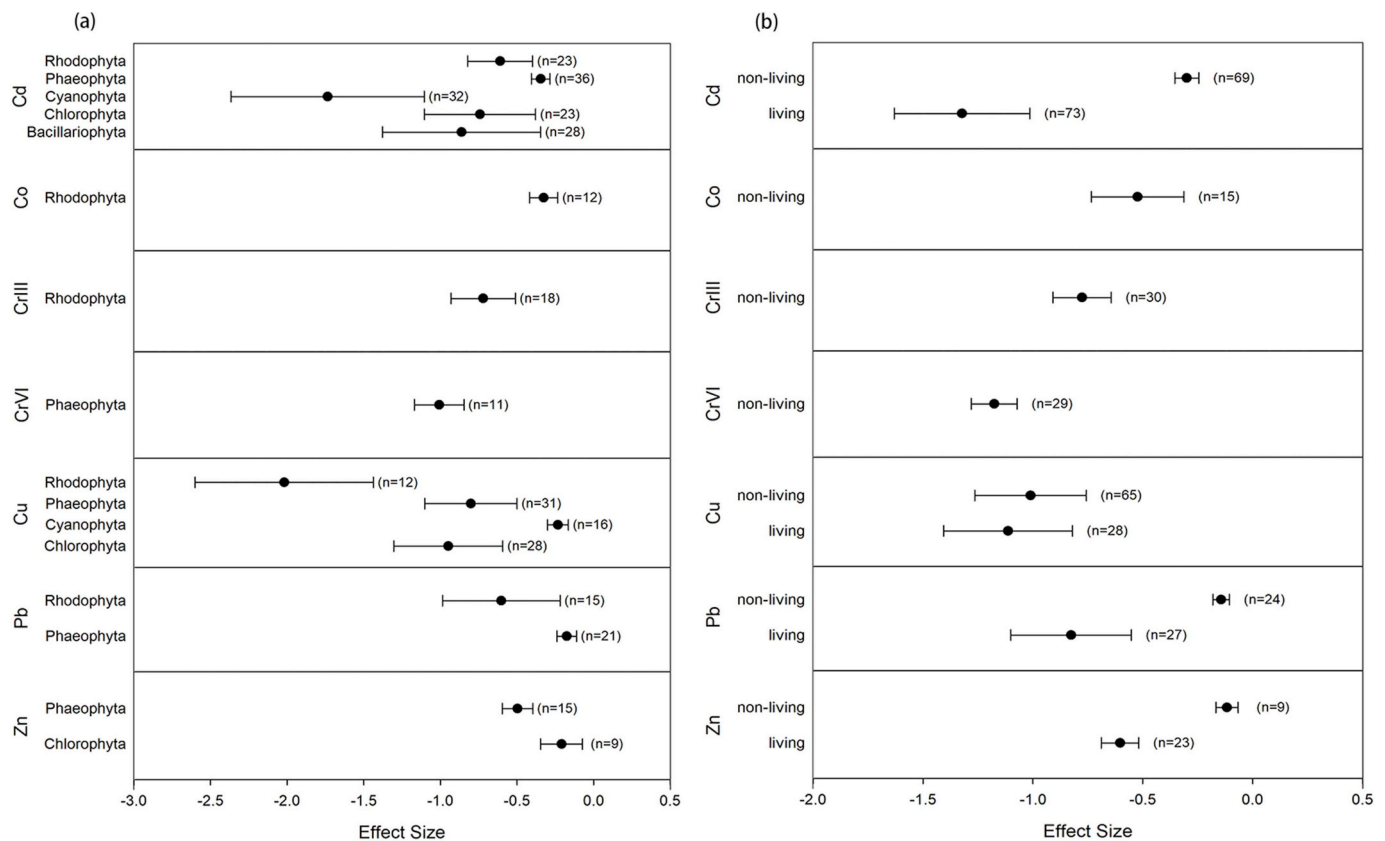


Fig. 2. (a) Effects of different algal phyla on heavy metal adsorption. (b) Effects of different algal states on heavy metal adsorption. Effect size processed by Ln, and error bars represent $\pm$ 95% confidence interval; n represents number of samples for each indicator. Data were tested via heterogeneity (Q), $P < 0.05$.

applications for heavy metal adsorption.

3.4. Algal state

The algal state (i.e., non-living or living) affected heavy metal adsorption. This study included six heavy metals (Cd, Cr, Co, Pb, Cu, and Zn); Cd had 37 data points, Cr^{3+} had 10, Cr^{6+} had 7, Co had 5, Pb had 17, Cu had 26, and Zn had 10 (with a total of 112 data points). This study found that non-living algae adsorbed heavy metals more efficiently than living algae did (Fig. 2(b)), which is consistent with other findings (Sandau et al., 1996; Kratochvil and Volesky, 1998; Jalali et al., 2002; Deng et al., 2009). A possible reason is that living algae are tolerant to heavy metals, or manifest toxic effects or altered physiological characteristics (Monteiro et al., 2011).

Fig. 2(b) shows that the order of heavy metal adsorption efficiency of non-living algae was $\text{Zn}^{2+} > \text{Pb}^{2+} > \text{Cd}^{2+} > \text{Co}^{2+} > \text{Cr}^{3+} > \text{Cu}^{2+} > \text{Cr}^{6+}$. The heavy metal adsorption efficiency of living algae was as follows: $\text{Zn}^{2+} > \text{Pb}^{2+} > \text{Cu}^{2+} > \text{Cd}^{2+}$; Co^{2+} , Cr^{3+} , and Cr^{6+} had insufficient data, because the number of data available for analyzing the adsorption efficiency of living algae was less than the minimum level required for meta-analysis. For Zn^{2+} , Pb^{2+} , and Cd^{2+} , the adsorption efficiency of non-living algae (ES values for Zn^{2+} , Pb^{2+} , and Cd^{2+} of -0.12 , -0.14 , and -0.30 , respectively) was much higher than that of living algae (ES values for Zn^{2+} , Pb^{2+} , Cd^{2+} of -0.60 , -0.83 , and -1.32 , respectively). However, for Cu^{2+} , the difference in adsorption efficiency between living algae (ES = -1.11) and non-living algae (ES = -1.01) was small.

The Cu^{2+} and Zn^{2+} adsorption efficiencies of living and non-living algae were low. This was likely because some binding modes of the algal functional groups during Cu^{2+} and Zn^{2+} adsorption were similar, such as the movement of C-N species in the amine group (Pavasant et al., 2006). The difference in Cu^{2+} and Zn^{2+} adsorption by algae is

reflected in the different binding affinities of different algal cell surfaces (Wilde et al., 2006) and environmental factors, e.g., freeze-drying or drying temperature (Matheickal and Yu, 1999; Ibrahim, 2011; Yang et al., 2015; Fawzy, 2016). Therefore, algal activity did not affect Cu^{2+} and Zn^{2+} adsorbance. On the other hand, non-living algae are unaffected by the environment and are thus more convenient and practical to use in the treatment of polluted water (Wilde et al., 2006). Differences in the Pb^{2+} and Cd^{2+} adsorption efficiencies of living and non-living algae were distinct. In particular, non-living algae had excellent Pb^{2+} adsorption properties.

Living algae are strongly affected by the environment. If heavy metal concentrations exceed tolerance thresholds, toxic effects on living algae influence their normal metabolic function, which, in turn, decreases their adsorption efficiency. Therefore, living algae should only be used to treat heavy metal concentrations of less than 10 mg/L, i.e., concentrations that do not exceed their tolerance threshold. Algal tolerance of heavy metals varies depending on the binding mechanism. In general, the tolerance is lower for living than for non-living algae because of heavy metal toxicity. Non-living algae mainly adsorb heavy metal ions through functional groups on their surface. During the process to produce algal powder, their particle size is usually reduced by grinding, which increases the surface area exposed to heavy metal ions. Thus, non-living algae are more frequently used as an adsorbent for industrial wastewater treatment (Mohamed, 2001).

Toxicological effects of heavy metals on algal adsorption efficiencies are not well understood. It is believed that heavy metals can affect the adsorption and detoxification of living algae by either blocking the surface channels of the cell membranes or by causing structural problems in cells: Heavy metals, occurrence and toxicity for plants: a review. According to Wilde et al., (2006), Zn^{2+} toxicity mostly originates from extracellular (rather than intracellular) Zn^{2+} and is likely due to the effects on cell membranes. Even relatively low metal concentrations

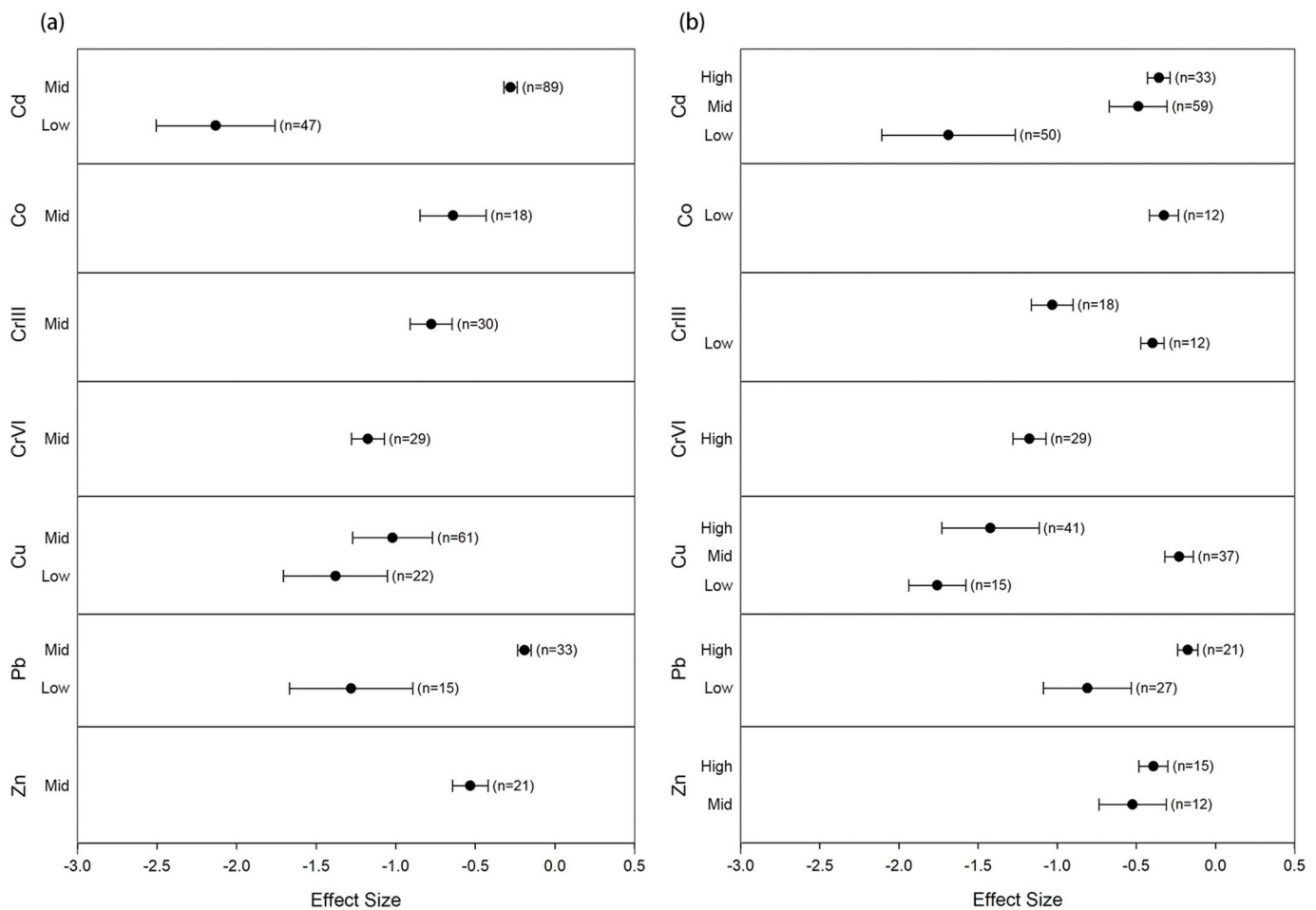


Fig. 3. (a) Effects of algal biomass on heavy metal adsorption. (b) Effects of different initial heavy metal concentrations on heavy metal adsorption by algae. Effect size was processed by Ln, and error bars represent $\pm$ 95% confidence interval; n represents number of samples for each indicator. In (a), “High” refers to biomass concentration of > 10 g/L (due to insufficient data, it is not reflected in the figure), “Mid” to biomass concentration of $1 \text{ g/L} \leq x \leq 10 \text{ g/L}$, and “Low” to biomass concentration of $< 1 \text{ g/L}$. In (b), “High” refers to initial heavy metal concentration of $> 100 \text{ mg/L}$, “Mid” to initial heavy metal concentration of $10 \text{ mg/L} \leq x \leq 100 \text{ mg/L}$, and “Low” to an initial heavy metal concentration of $< 10 \text{ mg/L}$. Data in graph were tested via heterogeneity (Q), $P < 0.05$.

may cause severe toxicity (Tchounwou et al., 2012), resulting in different adsorption capacities of living and non-living algae.

3.5. Algal biomass

Algal biomass (high biomass concentration: $> 10 \text{ g/L}$; moderate biomass concentration: $1\text{--}10 \text{ g/L}$; low biomass concentration: $< 1 \text{ g/L}$) impacted heavy metal adsorption. This study included six different heavy metals (Cd, Cr, Co, Pb, Cu, and Zn); Cd had 35 data points, Cr^{3+} had 10, Cr^{6+} had 7, Co had 6, Pb had 16, Cu had 24, and Zn had 7 (total of 105 data points).

As seen in Fig. 3(a), the adsorption efficiency of moderate algal biomass was greater than that of low algal biomass. This corroborates the results of previous studies. Wang and Chen, (2006) showed that the input of algal powder at a concentration of 4 g/L enhanced adsorption while minimizing the amount of required algal powder. The relationship between algal concentration and heavy metal adsorption efficiency indicated that the latter increased with increasing algal concentration before reaching a maximum level and subsequently decreasing. Hamdy, (2000) showed that within an algal biomass concentration range of $5\text{--}30 \text{ mg/L}$, adsorption efficiency increased sharply with increasing biomass concentration.

Similar trends were observed in the present study for Cd^{2+} and Pb^{2+} , with the highest adsorption efficiency having been achieved at an algal biomass concentration of $1\text{--}10 \text{ g/L}$ (with Cd^{2+} and Pb^{2+} ES

values of -0.28 and -0.19 , respectively). The adsorption efficiency did not significantly increase at algal concentrations above 10 g/L and was relatively low at algal concentrations below 1 g/L . The Zn^{2+} adsorption efficiency was also high (ES = -0.53 ; Fig. 3(a)). For Cu^{2+} ions, the adsorption efficiency of algal biomass at concentrations of $1\text{--}10 \text{ g/L}$ (ES = -1.02) was greater than that at concentrations of less than 1 g/L (ES = -1.38). This was also consistent with the results of previous studies: According to Wang and Chen, (2006), when the algal biomass concentration was low, metal ions in solution were not only adsorbed onto biomass surfaces but also entered the intracellular region, resulting in a concentration gradient. According to Romera et al., (2007), when the biomass concentration was high (e.g., $> 10 \text{ g/L}$), a shell effect occurred, and the protective active sites were not occupied by metal ions. Therefore, the excessive use of algae does not improve adsorption efficiency and may have an adverse effect on heavy metal adsorption (Ghoneim et al., 2014).

3.6. Initial heavy metal concentration

Heavy metal concentration (high: $> 100 \text{ mg/L}$; moderate: $10\text{--}100 \text{ mg/L}$; low: $< 10 \text{ mg/L}$) also impacted algal adsorption efficiency. This study included six heavy metals (Cd, Cr, Co, Pb, Cu, and Zn); Cd had 37 data points, Cr^{3+} had 10, Cr^{6+} had 7, Co had 4, Pb had 16, Cu had 26, and Zn had 9 (total of 109 data points). When the initial heavy metal concentration was $10\text{--}100 \text{ mg/L}$ or higher, algae had a

relatively high adsorption efficiency (Fig. 3(b)), which was similar to the results of previous studies (Karthikeyan et al., 2007; Lee and Chang, 2011). According to Wang and Chen, (2006), if the algal biomass remained unchanged, the metal ion adsorption rate increased with increasing initial ion concentration. Therefore, when the initial heavy metal concentration was lower than 10 mg/L, the maximum adsorption efficiency was relatively low. The algal adsorption efficiency was affected by heavy metal concentration (Fig. 3(b)). The trends observed for Cd^{2+} (ES = -0.36) and Pb^{2+} (ES = -0.17) were similar, and the highest adsorption efficiency was achieved at initial heavy metal concentrations of greater than 10 mg/L. In contrast, the algal adsorption effect was better when the concentrations of Co^{2+} (ES = -0.33) and Cr^{3+} (ES = -0.40) were less than 10 mg/L; Cu^{2+} adsorption performance improved at concentrations of 10–100 mg/L.

The possible reason for this is that different heavy metals have different toxicity thresholds. Before reaching the threshold, the adsorption rate mostly increased with increasing initial heavy metal concentration. However, after the threshold was exceeded, its magnitude decreased. Another possible cause of algal adsorption efficiency being affected by heavy metal concentration is the highest adsorption limit of algae. When the peak was reached, the algal adsorption efficiency gradually decreased. Non-living algae adsorbed heavy metal ions and bound them only in groups. After the adsorbed amount of metal ions reached a maximum, adsorption ceased. In addition to the binding ability of surface groups, living algae could also adsorb certain heavy metal ions in their vacuoles through biometabolism. However, excess heavy metals may have toxic effects and hence reduce the adsorption capacity (Sheng et al., 2007).

In summary, the adsorption of heavy metal ions by algae was limited at low initial metal ion concentrations. The heavy metal adsorption efficiency of algae increased with increasing concentrations to a certain limit, after which it gradually decreased. If the initial heavy metal concentration exceeded the algal adsorption capacity, the adsorption efficiency decreased with increasing heavy metal ion concentration.

3.7. Temperature

Temperature (high: > 28 °C; mid: 22–28 °C; low: < 22 °C) had little effect on algal adsorption efficiency. This study included six different heavy metals (Cd, Cr, Co, Pb, Cu, and Zn); Cd had 35 data points, Cr^{3+} had 10, Cr^{6+} had 6, Co had 5, Pb had 17, Cu had 25, and Zn had 8 (total of 106 data points; Fig. 4(a)).

The effect of temperature on the adsorption of heavy metals by algae was relatively small. At room temperature (near 25 °C) conditions, algal adsorption efficiency depended on adsorption capacity rather than temperature (Fig. 4(a)). Analyses of the collected data revealed that temperature variations between the studies were mostly only a few degrees Celsius, and most of the experiments were conducted at room temperature (Chaisuksant, 2003; Cruz et al., 2004; Lodeiro et al., 2004; Ibrahim, 2011; Yang et al., 2015). Algae usually exhibited relatively high efficiency at room temperature, but the adsorption of each heavy metal was affected differently.

Since large amounts of non-living algae were used in the experiments, a room temperature of 25 °C was typically used. For living algae, an optimum temperature range of 22–28 °C was mainly used. According to several studies, the biosorption capacity of algae was generally affected by temperature. The adsorption efficiency of different algal species has been demonstrated to depend on a variety of temperatures. At temperatures of 25–45 °C, the adsorption efficiency of some algae increased with increasing temperature (Verma et al., 2017), whereas for other species, either the opposite trend was observed (Uluozlu et al., 2010) or the adsorption efficiency did not change significantly with increasing temperature (Tüzün et al., 2005; Vilar et al., 2005). In some cases, the adsorption efficiency first peaked and then decreased (Zer et al., 1999), which was caused by interactions between the active binding sites of algal cell membranes and the adjacent molecules of the

adsorbed phase (Sarı and Tuzen, 2009).

3.8. Time

Time (long: > 24 h; mid: 3–24 h; short: < 3 h) influenced algal adsorption efficiency. This study included six different heavy metals (Cd, Cr, Co, Pb, Cu, and Zn); Cd had 37 data points, Cr^{3+} had 10, Cr^{6+} had 7, Co had 6, Pb had 17, Cu had 26, and Zn had 8 (total of 111 data points). In general, most algae achieved relatively high adsorption efficiencies over a short time period (< 3 h of contact between the algae and the medium). The difference in adsorption efficiency over 3–24 h depended on the metal (Fig. 4(b)). These results were similar to the findings of other studies: According to Hamdy, (2000), algae could adsorb a large number of heavy metal ions in a relatively short time under various conditions, after which the adsorption rate decreased sharply while the adsorption process reached equilibrium. In general, algae could adsorb more than 90% of heavy metal ions within 20 min of contact with the medium. The rapid adsorption that occurred during the early stage of contact between the algae and the water corresponded to physical adsorption, which accounted for 80–90% of the total adsorbed amount. Slow adsorption at the later stage was generally related to the energy-driven metabolism of living algae, which had a low adsorption rate. Usually, after 3 h, the algal adsorption rate did not increase (Sarı and Tuzen, 2008; Jacinto et al., 2009).

As shown in Fig. 4(b), the algal adsorption efficiencies for Cd^{2+} (with ES values for long, mid, and short periods of -2.22, -0.72, and -0.27, respectively) and Pb^{2+} (with ES values for long and short periods of -1.28 and -0.18, respectively) decreased with increasing adsorption time. The adsorption efficiency of Cu^{2+} (with ES values of -1.50, -1.82, and -0.24 for long, mid, and short periods, respectively) was the lowest in mid time period (3–24 h) and increased after 24 h. The difference between Zn^{2+} (with ES values for long and short periods of -0.48 and -0.45, respectively) and Cr^{6+} (with ES values for mid and short periods of -1.11 and -1.23, respectively) was small (Fig. 4(b)). In addition, the adsorption efficiency of Zn^{2+} and Cr^{6+} changed little over time. The reason for this may be that excessive adsorption time led to the poisoning of living algae, which affected adsorption efficiency. Algal adsorption efficiency may decrease when adsorption time exceeds 24 h (Saleh, 2015).

3.9. pH value

The pH value (low: < 6.5; mid: 6.5–7.5; high: > 7.5) influenced algal adsorption efficiency. This study included six different heavy metals (Cd, Cr, Co, Pb, Cu, and Zn); Cd had 34 data points, Cr^{3+} had 10, Cr^{6+} had 7, Co had 5, Pb had 16, Cu had 23, and Zn had 8 (total of 103 data points). The number of data relevant to alkaline conditions was too small, and the majority of data points were excluded. The data for neutral conditions that met the minimum standard for meta-analysis were included (Fig. 5).

Algae generally had higher adsorption efficiencies for heavy metal ions under neutral and weakly acidic conditions (Fig. 5). When the pH was < 7.5, the probability of reaching the highest adsorption efficiency was relatively high, and the experimental adsorption efficiency was indeed high. As the pH increased, the adsorption efficiency of most algal species drastically declined (Zhou et al., 1998). This phenomenon could be explained by the precipitation of metal ions, such as Cu^{2+} and Zn^{2+} , in an alkaline environment (Montazer-Rahmati et al., 2011) and the associated adverse effects on algae. Another reason why algae adsorb heavy metals at pH values in the range of 4–7.5 may be the competitive adsorption of hydrogen and heavy metal ions. The pH can affect the binding strength of heavy metal ions by affecting the formation and activity of certain functional groups (Prakash and Kumar, 2013). Therefore, a sudden increase or decrease in adsorption efficiency may occur at a pH of 4 (which is the approximate pH of industrial wastewaters). First, the number of hydrogen ions in the solution is high

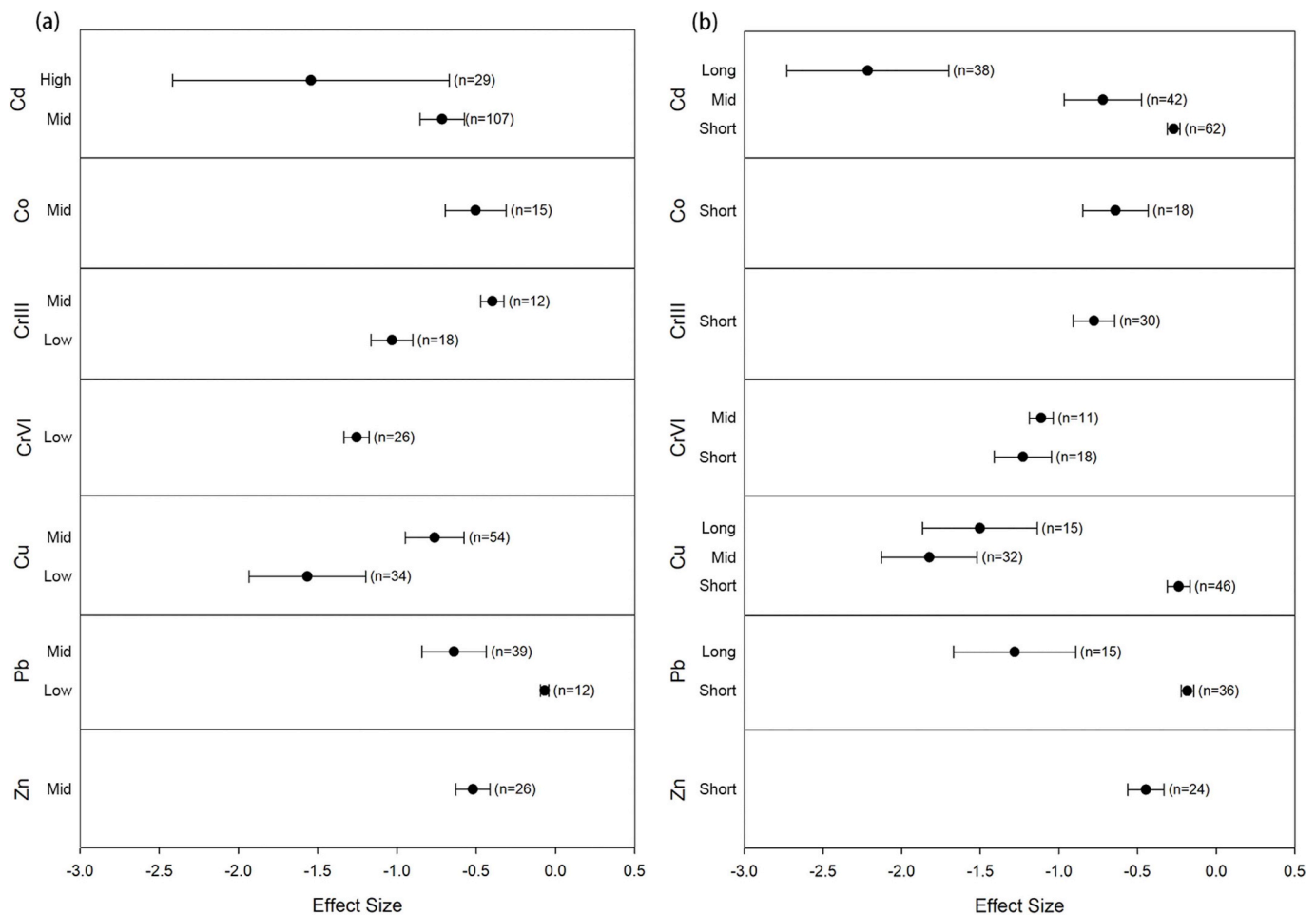


Fig. 4. Effects of different (a) temperatures and (b) time on heavy metal adsorption by algae. Effect size was processed by Ln, and error bars represent $\pm 95\%$ confidence interval; n represents number of samples for each indicator. In (a), “High” refers to temperature of $> 28^\circ\text{C}$, “Mid” to temperature of $22^\circ\text{C} \leq x \leq 28^\circ\text{C}$, and “Low” to temperature of $< 22^\circ\text{C}$. In (b), “Long” refers to time of $> 24\text{ h}$, “Mid” to time of $3\text{ h} \leq x \leq 24\text{ h}$, and “Short” to time of $< 3\text{ h}$. Data in graph were tested by heterogeneity (Q), $P < 0.05$.

enough to compete with heavy metal ions for algal adsorption sites. Second, some metal ions precipitate at this pH (Kuyucak and Volesky, 1989). This should be considered during the practical application of algae for treating heavy metal-contaminated wastewater, and algal species that are more resistant to acidic environments should be used (Cossich et al., 2003; Han et al., 2005).

In general, different algae have different tolerances to pH, and optimal pH values mostly ranged from 4 to 7.5 (Liu et al., 2009). A weakly acidic or neutral environment was suitable for most algal species to adsorb Cd^{2+} , Cr^{3+} , and Cr^{6+} , whereas the adsorption of Pb^{2+} and Cu^{2+} occurred in more acidic environments. For Zn^{2+} adsorption, a neutral pH environment was necessary. When multiple heavy metal contaminants are present, one or two types of algae should be selected for adsorption based on the pH of the aqueous solution, and the adsorption efficiency could be improved without increasing the amount of algae and associated cost.

4. Conclusions and outlook

4.1. Conclusion

Using a meta-analysis, this study determined the differences in heavy metal adsorption by algae and analyzed various factors that resulted in different adsorption efficiencies. The following conclusions were drawn:

1. Considering various environmental factors, the heavy metal adsorption efficiency of algae was ranked as Phaeophyta $>$ Rhodophyta $>$ Chlorophyta $>$ Cyanophyta $>$ Bacillariophyta. In addition, for Cu^{2+} -polluted water, the adsorption efficiency of Cyanophyta was higher and had greater advantages.
2. Algal activity affected adsorption efficiency. Generally, the adsorption efficiency of non-living algae was higher than that of living algae. However, algal adsorption performance depended on the type of heavy metal. Non-living algae were more suitable for heavy metal adsorption because of their small volume, large surface area, and high number of surface functional groups. Moreover, heavy metal toxicity was not problematic for non-living algae, and the adsorption efficiency was similar in different environments.
3. Changes in algal biomass and heavy metal concentration affected algal adsorption efficiency. When algal biomass and heavy metal concentration were 1–10 g/L and 10–100 mg/L, respectively, the adsorption efficiency was the highest.
4. Temperature did not affect the heavy metal adsorption efficiency of algae, but time and pH did. Algae adsorbed heavy metals over a short period, and a maximum adsorption capacity of 80–90% could be achieved within 20 min of contact between the algae and medium. In weakly acidic or neutral environments ($\text{pH} = 4\text{--}7.5$), the adsorption efficiency was high, and the pH value of industrial wastewater is usually within this range. Therefore, algae have considerable advantages in sewage treatment.

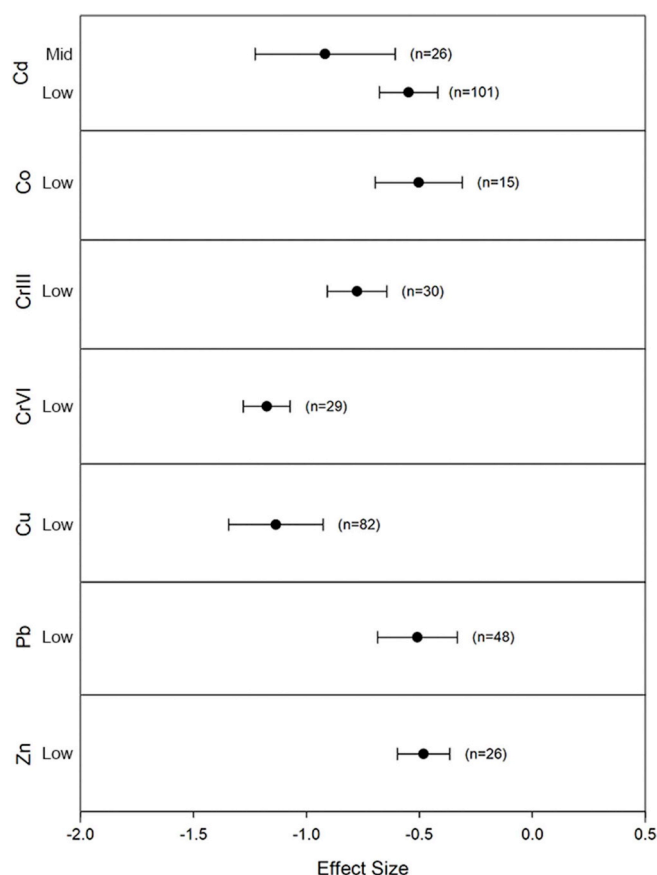


Fig. 5. Effects of pH value on algal heavy metal adsorption. Effect size was processed by Ln, and error bars represent $\pm$ 95% confidence interval; n represents number of samples for each indicator. “Mid” refers to pH of $6.5 \leq x \leq 7.5$, and “Low” to pH of < 6.5 . Data were tested by heterogeneity (Q), $P < 0.05$.

4.2. Outlook

Although the use of algae to purify heavy metal-contaminated water bodies still has some shortcomings in terms of practical application, this technique has the advantages of being rapid, having high adsorption efficiency, reducing pollution, and providing environmental protection. Algal adsorption as a solution to heavy metal pollution therefore has excellent application potential and has been widely researched. Even though research has advanced in terms of both the mechanism and application of algal heavy metal adsorption, many gaps in knowledge remain. For example, whether the approach can enhance heavy metal adsorption by combining different algal species still needs to be determined. Further studies are also needed to assess whether algae that exhibit a strong adsorption capability can effectively adsorb heavy metals in practical applications as well as to develop effective methods to reduce the manufacturing costs of non-living algae for large-scale applications. Solutions to these problems will further contribute to a robust scientific basis for future treatment of heavy metal-contaminated water.

Author contribution

Zeyu Lin: conception, resources, data curation, writing; **Jing Li:** methodology, software, validation, project administration; **Yaning Luan:** conceptualization, supervision, funding acquisition; **Dai Wei:** Methodology, Software, conceptualization, supervision.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ecoenv.2019.110089>.

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